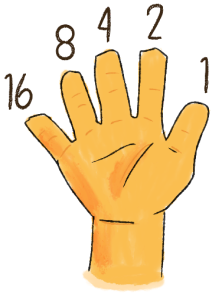
Count on Fingers

Grade 7

Introduction

We usually use our fingers to count from 1 to 10. But is that the maximum we can count with our fingers? We know that numbers can be represented in the powers of 10. Though this is not the only way to represent the number.



In this activity, students will explore a surprising idea: using fingers to count much higher numbers. They will discover that with just 10 fingers, they can count to numbers more than 1000. With 33 fingers, we can represent the human population of the world. They will understand binary representation and the powers of 2 through finger counting. This connects directly to how computers store information using only 0s and 1s. By the end of this activity, students

Activity	Time	Description
Launch	5 min	Ask students: How high can you count using the fingers of both hands? Expected responses: 10 → Unary counting 25 → Grouping method (5×5) 30 → Finger segments method Demonstrate these methods briefly. Then say: What if I told you we can count beyond 1000 using just 10 fingers? Video Reference: https://youtu.be/RT6j8Bvs1HY?si=pNOrXf89LtOk_Vdt

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Explore	20 min	Students work alongside the worksheet while teacher guides discussion. Student Worksheet: https://docs.google.com/document/d/1LGj23wKJ0dyiS1lrZ1hhyLxz3gRk11A68--dbgAEP5Y/edit?usp=sharing
Discussions & Conclusion	10 min	Discussing the worksheets. With binary representation, we can represent any numbers efficiently.

CT Components

Algorithmic Thinking
We follow a step-by-step method to represent numbers using fingers. Also, we used a step-by-step approach to convert a decimal number to binary.

Decomposition
Breaking down the conversion of the number to binary into one bit at a time.
Breaking the numbers into sums of powers of 2.

Pattern Recognition
Recognising doubling pattern: 1, 2, 4, 8, 16…

Abstraction
Using fingers to count and to represent the binary number system.

Logic
Every finger can correspond to a power of 2, and every number can be represented in the binary system with the fingers.

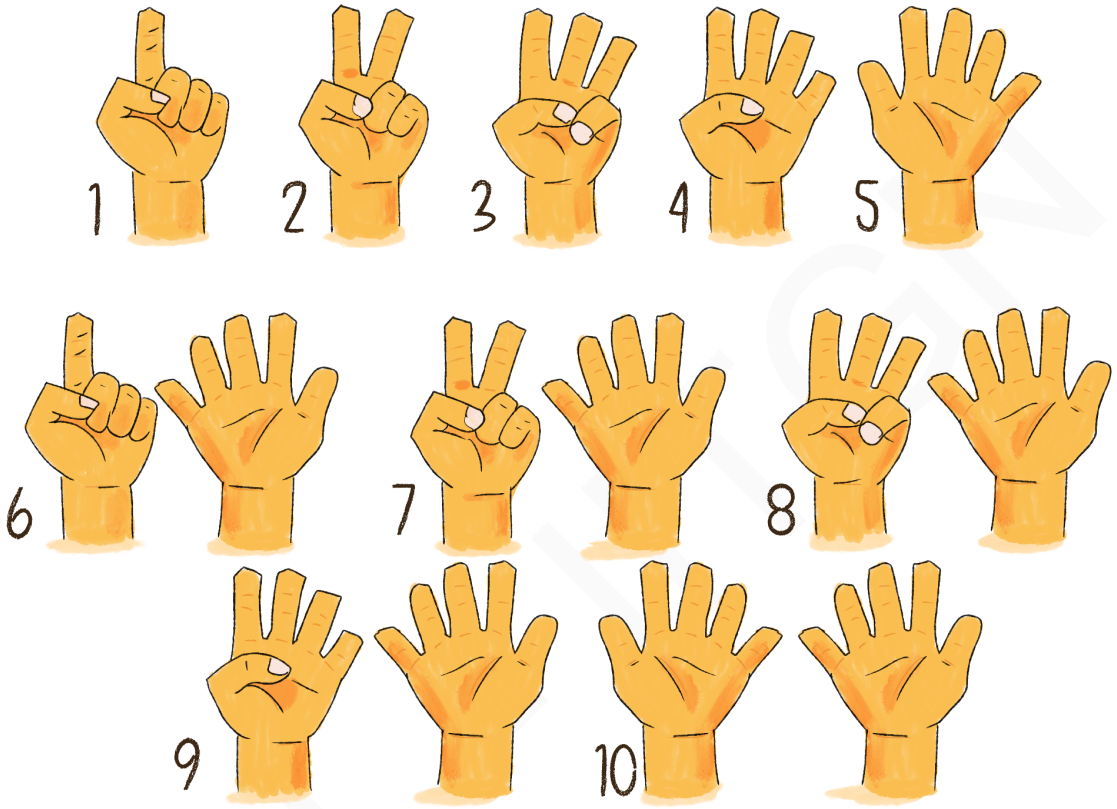
Generalisation
Recognising a doubling pattern of 1, 2, 4, and so on, and adding the corresponding powers gives us the binary representation; we conclude that any number can be represented as a sum of powers of 2. Understanding that with n fingers, the maximum count is: 2^n − 1.

Evaluation
Checking whether a finger combination correctly represents a number.

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Activity: Count on Fingers

From an early age, we rely on the ability to count. Children learn to count with their fingers at an early age. The Latin word for fingers, “digitus”, is the root of the English word digit. Using one finger, we can count up to 5 numbers. No finger lifted up means it represents 0. Similarly, using fingers from other hands, we will be able to count 5 more, i.e., 10.



We usually use both hands' fingers to count from 1 to 10. But is that the maximum we can count with our fingers?

This is known as the unary representation of a number. To represent a number N, a symbol representing 1 is repeated N times. Here, N fingers will be required to represent N.

Q. Using the above method, how many fingers are required to count up to 15?

a) 5
b) 10
c) 15
d) 20

Answer c) 15

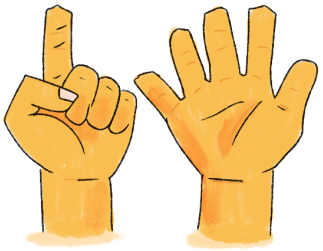
To represent numbers upto 15, 15 fingers will be required.

CT Competency: Abstraction, Evaluation, Logic

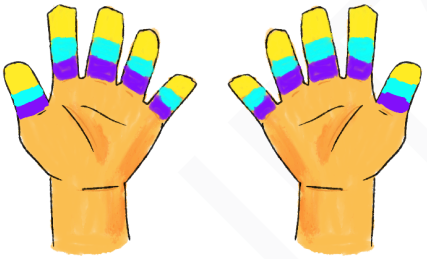
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It is the simplest way to represent a number. It is also the oldest, dating back to the stone age and has been used ever since. This limits the counting of the number with fingers of both the hands to 10.

Now, we will use the second hand's finger as a counter. After every 5 counts on one hand, a finger is raised on the other hand. With that, we can count up to 25.



A slightly more advanced method: if you divide your fingers into three sections, double that for two hands, you can go up to 30.



But that will still be 30 only. How should we count so that we reach numbers greater than 1000 ?

There is a way to count up to a number more than 1000.

For this, we are going to define a placement logic of finger as Down(0) and Up (1)

Image	Placement	Count
	Down	0
	Up	1

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And will assign numbers to the fingers as shown,



Q. What patterns could you observe with the numbers on fingers?

a) All numbers are powers of 2
b) All numbers are less than 10
c) All numbers are powers of 3
d) Numbers are not related to each other

Answer

b) All numbers are Powers of 2

These numbers double each time. They are called powers of two.

CT Competency: Pattern Recognition, Abstraction, Logic

Each finger on your hand represents a number, and we add them up when the finger is raised.

In the shown example, all fingers are raised up except the thumb.



So we need to add +16 for the little finger, +8 for the ring finger, +4 for the middle finger, +2 for the index finger, but +0 for the thumb. 16 + 8 + 4 + 2 = 30, which makes it number 30.

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The following table summarises the equivalent of every decimal number with the corresponding fingers.

Decimal Numbers	**Made by**					
1	1					up
2	2				up	
3	2+1				up	up
4	4			up		
5	4+1			up		up
6	4+2			up	up	
7	4+2+1			up	up	up
8	8		up			
9	8+1		up			up
10	8+2		up		up	
11	8+2+1		up		up	up
12	8+4		up	up		
13	8+4+1		up	up		up
14	8+4+2		up	up	up	
15	8+4+2+1		up	up	up	up
16	16	up				
17	16+1	up				up
	and so on....					

Q. What decimal number is represented by the given binary fingers?



a) 25
b) 26
c) 24
d) 23

Answer
b) 26

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Here, we need to add +16 for the little finger, +8 for the ring finger, and +2 for the index finger, giving a total of 26.
CT Competency: Decomposition, Algorithmic Thinking, Logic, Abstraction

Q. Which is the highest decimal number that can be represented if all the fingers are raised up on one hand?



a) 16
b) 30
c) 32
d) 31

Answer
d) 31
With one hand, when all the fingers are raised, we can count +16 for the little finger, +8 for the ring finger, +4 for the middle finger, +2 for the index finger, +1 for the thumb. 16+8+4+2+1 = 31.
That means the highest decimal number that can be represented by the fingers of one hand is 31.
CT Competency: Generalisation, Abstraction, Logic, Evaluation

This is known as the Binary number system, often known as the language of computers. In our day-to-day life, we use the decimal number system, which uses the numbers 0, 1, 2, 3, 4, 5, 6, 7, 8, and 9. The decimal number system is based on bundles of ten (Powers of 10, which is 1,10,100,1000, 1000,...). Similarly, the binary number system is based on bundles of two (Power of 2, which is 1,2,4,8,...). So if we compare the two numbering systems for place values, this is what it looks like….

	Decimal (Bundles of Ten)	Binary (Bundles of Two)
Units Place	1	1
First Bundle	10 = 10	2 = 2
Second Bundle	10x10 = 100	2x2 = 4
Third Bundle	10x10x10 = 1000	2x2x2 = 8

The binary number system (popularly known as the language of computers) deals with only two numbers: 0 and 1. We can convert any decimal number to its binary equivalent and vice versa. We can convert any decimal number into a binary representation by writing it with the power of 2.

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For example, 23 = 16 + 4 + 2 + 1. Show this to the students.

23 in decimal	$10^1 = 10$	$10^0 = 1$
23	2	3

23 in binary	$2^4 = 16$	$2^3 = 8$	$2^2 = 4$	$2^1 = 2$	$2^0 = 1$
$(10111)_2$	1	0	1	1	1

We can also do this by an algorithm, which is presented below (which teacher may present to the student):

i. Divide the decimal number by the base (2) of the binary number.
ii. Note the remainder from the first step till the end of the division process.
iii. The remainders obtained as 0 or 1 are written in order from the last step to the first step of division.

Let's find out the binary equivalent of the decimal number 11.

Divisor	Decimal Number	Corresponding Remainder
2	11	1
2	5	1
2	2	0
	1	

The highlighted cells represent the binary representation, written from bottom to top. So, 11 = $(1011)_2$. The binary number 1011 can be changed to its corresponding decimal number as:

A	B	C	D
2^3 x 1	2^2 x 0	2^1 x 1	2^0 x 1

Decimal number = A+B+C+D = 1 x 2^3 + 0 x 2^2 + 1 x 2^1 + 1 x 2^0

= (1x8)+ (0 x 4) + (1 x 2) + (1 x 1) =8+2+1 = 11

Q.What will be the binary representation for the decimal number 14?

a) 1010
b) 1100
c) 1110
d) 1111

Answer

c) 1110

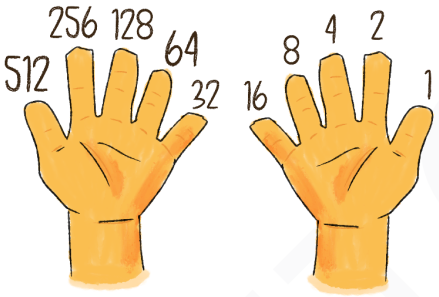
Decimal 14 can be written in power of 2s as 8 + 4 + 2 = 1 x 2^3 + 1 x 2^2 + 1 x 2^1 + 0 x 2^0 = 1110

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CT Competency: Algorithmic Thinking, Decomposition, Generalisation

The number of digits or bits (**bi**nary digi**ts**) in a binary number depends on the number. For example, numbers up to 7 can be represented in a 3-bit binary number, numbers from 8 to 15 in a 4-bit binary number, numbers from 16 to 31 similarly need an additional bit, hence a 5-bit binary number, and so on. Here, a bit is represented by a finger. Raised fingers are taken as 1, and fingers that have not been raised are considered as 0.

The same is also valid for both hands, where you count the fingers from left to right in the following way:



Q. What decimal number is represented by the given binary fingers?

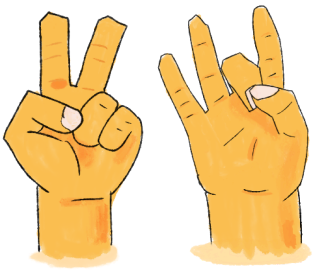


a) 32
b) 33
c) 64
d) 65

Answer
a) 32
Only the little finger of the second hand is raised, representing 32.
CT Competency: Algorithmic Thinking, Pattern Recognition, Logic, Abstraction

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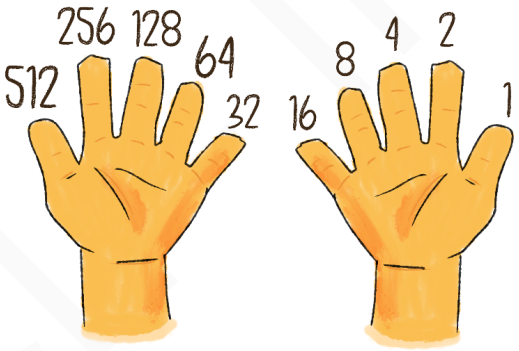
Q. What decimal number is represented by the given binary fingers?



a) 346
b) 406
c) 410
d) 411

Answer
C) 410
Adding the assigned value of the raised fingers 256+128+16+8+2 = 410
CT Competency: Algorithmic Thinking, Pattern Recognition, Logic, Abstraction

When all fingers are up on both hands, we can represent decimal numbers up to 1023.



1023 = 512+256+128+64+32+16+8+4+2+1, though we can count a total of 1024 numbers, counting a fist as 0. With this binary representation, we can count past 1,000 using only 10 fingers.

By recognising a doubling pattern of 1, 2, 4, and so on, and adding the corresponding powers, we can obtain the binary representation; we conclude that any number can be represented as a sum of powers of 2. Understanding that with n fingers, the maximum count is: 2^n − 1.

This binary concept has its roots in the study of poetry in ancient India, which was well studied by Acharya Pingala's and documented in his work Chandaḥśāstra (~2 BCE).

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